**Cohort Exercise: Deriving Tests from the Sequence Diagram**

Based on the Sequence Diagram in the folder, complete the following test design tasks:

**Task 1: Identify Units and Design Unit Tests (10pts)** Look at your lifelines. A "unit" typically represents a single class or component in your architecture (e.g., the UI, the Controller, the Service).

1. **Identify the units** in your Checkout sequence diagram.
2. **Write** 2 **unit test cases** (usually representing the success and failure paths) for **each** identified unit.
3. For every testcase, you must explicitly define:
   - **Target Unit:** (The Subject Under Test)
   - **Test Name/Scenario:**
   - **Inputs:** (What data the test passes to the unit)
   - **Expected Outputs:** (What the unit should return or state it should change)
   - **Mocked Input/Output pairs:** (Which dependencies are mocked, what fake data they receive, and what fake data they return)

**Task 2: Boundary Value Analysis (BVA) Test Suite (5pts)** Pick **one** of the specific unit inputs from Task 1 that has natural constraints (for example: the credit card string length, or an implied order total amount).

1. Identify the input variable and define its valid domain/range (e.g., a credit card must be exactly 16 digits).
2. Derive a **Robust Boundary Value Test Suite** for this variable.

**Task 3: Bottom-Up Integration Testing (10pts)**

1. Using the units you identified in Task 1, design an integration testing plan.
2. State explicitly what is *real* and what is *mocked* at each conceptual step of your bottom-up integration.

**Task 4: Deriving End-to-End (E2E) Test Cases (5pts)** End-to-End tests simulate the real user (`Student`) traversing the entire system, from the Frontend UI all the way to the Database.

1. Discuss how you can use the pathways in your Sequence Diagram to derive your E2E test cases.
2. Specifically, how do the **Combined Fragments** (`alt/else` blocks) in your diagram dictate the total number of E2E tests you need to write for the checkout flow? Briefly define the exact starting actions and final user-facing results for each.

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